Jahangirnagar University **T1**
Department of Computer Science and Engineering
Part II Semester II B. Sc (Honors) Examination - 2021
Course Name: Introduction to Probability and Statistics
Course Number: STAT-251

Time: 01 Hour **Marks: 30**

(Answer the following Questions. Each Question carries equal Marks.)

Q1. The accompanying frequency distribution of wind speed (in knots) of a particular location is generated by the researcher. Find the (i) mode, (ii) Karl Pearson's coefficient of Kurtosis; and (iii) Comment on your result.

Class Interval	Frequency
0.00 - 0.89	63
0.89 - 1.78	128
1.78 - 2.67	209
2.67 - 3.56	106
3.56 - 4.45	76
4.45 - 5.35	41
5.35 - 6.24	19
6.24 <	12

(Marks for Q1: 4+4+2=10)

Q2. The following data give the time (in seconds) that each of thirty two students selected from a university waited in line at their bookstore to pay for their textbooks in the beginning of the Fall 2018 semester: 45, 17, 29, 28, 25, 45, 24, 26, 5, 24, 27, 29, 29, 35, 8, 53, 1, 23, 40, 1, 37, 25, 28, 28, 3, 30, 25, 60, 4, 29, 53, and 26.

(i) Prepare a box-and-whisker plot. (ii) Are these data skewed in any direction? (iii) Find the mild and extreme outliers. (iv) Also, find the portion of the outliers in each category from the data set.

(Marks for Q2: 4+2+2+2=10)

Q3. The article "Characterization of Highway Runoff in Austin, Texas, Area" (*J. of Envir. Engr., 1998: 131–137*) gave a scatter plot, along with the least squares line, of X= rainfall volume (m^3) and Y=runoff volume (m^3) for a particular location. The accompanying values were read from the plot.

X:	5	12	14	17	23	30	40	47	55	67	72	81	96	112	127
Y:	4	10	13	15	15	25	27	46	38	46	53	70	82	99	100

(a) Fit a regression line to estimate runoff volume.

(b) Calculate an estimate of the runoff volume when the rainfall volume is 50.

(c) Evaluate the value of the correlation coefficient and coefficient of determination. Comment on your result.

(Marks for Q3: 4+2+4=10)

Best of Luck

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T3

Time: 75 Minutes **Marks: 30**

(Answer the following Questions. Each Question carries equal Marks.)

Q1. What is meant by Rank correlation? What are the differences with point biserial correlation? In what situation the correlations are used? **[6]**

Q2. In a study of the relationship between age and the EEG, data were collected on 20 subjects between ages 20 and 60 years. The following Table shows the age and a particular EEG output value for each of the 20 subjects. The investigator wishes to know that this particular EEG output is inversely correlated with age? **[6]**

ID	1	2	3	4	5	6	7	8	9	10
Age	20	21	22	24	27	30	31	33	35	38
Egg	98	75	95	100	99	65	64	70	85	74

ID	11	12	13	14	15	16	17	18	19	20
Age	40	42	44	46	48	51	53	55	58	60
Egg	68	66	71	62	69	54	63	52	67	55

Q3. What is meant by Gamma distribution? In what situation the Gamma distribution becomes to Exponential distribution. Write down some uses of Gamma distribution. **[6]**

Q4. The time of failure density for a small computer system has a Weibull density with location parameter $\gamma = 0$, shape parameter $\beta = 1/2$, and scale parameter $\delta = 100$. What fraction of these units will survive to 500 hours? What is the mean time of failure, and standard deviation of time of failure? **[6]**

Q5. Consider a historical record of the temperature of electronic devices and found the temperature of different days as below mean temperature, and above mean temperature: where 0 and 1 are considered for below mean temperature and above mean temperature respectively. The transition matrix is obtained as – **[6]**

$$TM = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} 224 & 96 \\ 96 & 234 \end{bmatrix} \end{array}$$

Find transition probability matrix, transition probability graph, steady state probability, mean recurrence time and first passage time $f_{01}^{(4)}$

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(Answer the following Questions. Each Question carries equal Marks.)

Q1. A bank operate both derive up facility and a walk-up window. On a randomly selected day X = the proportion of time that the derive up facility is in use (at least one customer is being served or waiting to be served) and Y = the proportion of time that the walk up window is in use. Then the set of possible values for (X,Y) is the rectangle $D=\{(x,y): 0\le x\le 2,\ 0\le y\le 3\}$. Suppose the joint density function of (X,Y) is given by

$$f_{XY}(x,y)=k(2x+y^2)\ ; \qquad 0\le x\le 2, \qquad 0\le y\le 3.$$

(a) Obtain the value of k in the *pdf*.

(b) What is the probability that neither facility is busy more than 2 of the time?

(c) Find $E[Y\mid X=1.5]$.

(d) Also, find the value of correlation coefficient, ρ_{XY}.

(Marks for Q1: 1+2+3+4=10)

Q2. A consumer organization that evaluates new automobiles customarily reports the number of major defects in each car examined. Let X denote the number of major defects in a randomly selected car of a certain type. The *cdf* of X is as follows (besides):

$$F(x)=\begin{cases}0 & x<0\\ 0.06 & 0\le x<1\\ 0.19 & 1\le x<2\\ 0.39 & 2\le x<3\\ 0.67 & 3\le x<4\\ 0.92 & 4\le x<5\\ 0.97 & 5\le x<6\\ 1.00 & 6\le x\end{cases}$$

(i) What is the pmf of X?

(ii) Find $P(2\le X<6)$, $P(X>4)$.

(iii) If $H(X)=3+2X^2$, estimate $V[H(X)]$.

(Marks for Q2: 2+4+4=10)

Q3. A worker has asked her supervisor for a letter of recommendation for a new job. She estimates that there is an 80 percent chance that she will get the job if she receives a strong recommendation, a 40 percent chance if she receives a moderately good recommendation, and a 10 percent chance if she receives a weak recommendation. She further estimates that the probabilities that the recommendation will be strong, moderate, and weak are 0.7, 0.2, and 0.1 respectively.

(a) How certain is she that she will receive the new job offer?

(b) Given that she does receive the offer, how likely should she feel that she received a strong recommendation; a moderate recommendation; a weak recommendation?

(c) Given that she does not receive the job offer, how likely should she feel that she received a strong recommendation?

(Marks for Q3: 2+6+2=10)

Best of Luck